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Department Of Artificial Intelligence & Data Science

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Date:- 20th July, 2026

LO:- LO-04

Experiment-02

AIM:- To Containerize Application using Docker

Theory:- Docker is an open-source containerization platform used to develop, package, and run applications inside lightweight, isolated environments called containers. It allows developers to bundle an application along with its dependencies, libraries, configuration files, and runtime environment into a single portable unit.

Docker solves the common problem of applications working differently across different environments by ensuring that the same container can run consistently on a developer machine, testing server, or production system.

Docker Commands:-

Check Docker version

Display Docker information

Download Docker image

List Docker images

Run a Docker container

List running containers

Docker images are built using a layer-based architecture.

Each image consists of multiple read-only layers. During the pull operation:

1. Docker checks which layers are already available locally.
2. It downloads only missing layers.
3. It verifies the integrity of each layer.
4. It stores the complete image locally.

This improves download speed and saves storage space.

RESULT:-

1. **Docker –version**

```
PS C:\Users\Gyaan> docker --version
Docker version 29.6.1, build 8900f1d
PS C:\Users\Gyaan> docker version
Client:
 Version:           29.6.1
 API version:       1.55
 Go version:        go1.26.4
 Git commit:        8900f1d
 Built:             Fri Jun 26 11:43:32 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.82.0 (233772)
 Engine:
  Version:          29.6.1
  API version:      1.55 (minimum version 1.40)
  Go version:       go1.26.4
  Git commit:       8ec5ab3
  Built:            Fri Jun 26 11:41:04 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.5
  GitCommit:        e53c7c1516c3b2bff98eb76f1f4117477e6f4e66
 runc:
  Version:          1.3.6
  GitCommit:        v1.3.6-0-g491b69ba
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

2. Docker Version

```
PS C:\Users\Gyaan> docker --version
Docker version 29.6.1, build 8900f1d
PS C:\Users\Gyaan> docker version
Client:
 Version:           29.6.1
 API version:       1.55
 Go version:        go1.26.4
 Git commit:        8900f1d
 Built:             Fri Jun 26 11:43:32 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

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  Built:            Fri Jun 26 11:41:04 2026
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  Version:          1.3.6
  GitCommit:        v1.3.6-0-g491b69ba
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

3. Docker Info

```
PS C:\Users\Gyaan> docker info
Client:
 Version:    29.6.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.98.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.27.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.35.0-desktop.2
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.3.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.4.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.5
    Path:     C:\Program Files\Docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
```

4. Docker Images

```
PS C:\Users\Gyaan> docker images

IMAGE                                    ID             DISK USAGE   CONTENT SIZE   EXTRA
nginx/nginx-prometheus-exporter:latest   bf76e58df548       19.7MB         5.24MB
nginx:latest                             5a88c9c45479        241MB           66MB
rancher/nginx:latest                     ddf6bbcc6f8a        805MB          239MB
PS C:\Users\Gyaan>
```

5. Docker Pull Ubuntu

```
PS C:\Users\Gyaan> docker pull ubuntu:24.04
24.04: Pulling from library/ubuntu
ca2678b20700: Pull complete
55378bbcdb75: Download complete
Digest: sha256:4fbb8e6a8395de5a7550b33509421a2bafbc0aab6c06ba2cef9ebffbc7092d90
Status: Downloaded newer image for ubuntu:24.04
docker.io/library/ubuntu:24.04
```

6. Docker Pull Ngix

```
PS C:\Users\Gyaan> docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
```

7. Docker ps

```
PS C:\Users\Gyaan> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\Gyaan> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\Gyaan>
```

8. Docker ps -a

```
PS C:\Users\Gyaan> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\Gyaan> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\Gyaan>
```

9. Docker run -it Ubuntu

```
PS C:\Users\Gyaan> docker run -it ubuntu
root@cfd7b2df4ff5:/#
```

10. Docker Image ls

```
PS C:\Users\Gyaan> docker image ls
                                                                    i Info →  U In Use
IMAGE                                      ID             DISK USAGE   CONTENT SIZE   EXTRA
nginx/nginx-prometheus-exporter:latest     bf76e58df548       19.7MB         5.24MB
nginx:latest                               5a88c9c45479        241MB           66MB
rancher/nginx:latest                       ddf6bbcc6f8a        805MB          239MB
ubuntu:24.04                               4fbb8e6a8395        119MB         31.7MB
ubuntu:latest                              3131b4cc82a7        161MB         45.3MB   U
PS C:\Users\Gyaan>
```

11. Docker Image prune

```
PS C:\Users\Gyaan> docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
PS C:\Users\Gyaan>
```

Conclusion:- Docker is a powerful containerization platform that enables developers to build, deploy, and manage applications efficiently using containers. Docker commands provide various functionalities such as creating, running, managing, and removing containers, handling images, managing networks and volumes, and automating multi-container applications using Docker Compose. By using Docker, applications can run consistently across different environments with improved portability, scalability, and resource utilization. Understanding basic Docker commands is essential for effective container management and is widely used in modern software development, DevOps, and cloud computing environments.